

Ex.No:1: Simulation of FDMA in MATLAB

Objective 1

Program

```
clc; clear; close all;

fs=1000; t=0:1/fs:1;

TBW=600; Users=3;

CBW=TBW/Users;

fc=[100 300 500];

m=sin(2*pi*10*t);

s1=m.*cos(2*pi*fc(1)*t);

s2=m.*cos(2*pi*fc(2)*t);

s3=m.*cos(2*pi*fc(3)*t);

subplot(2,1,1)

plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on

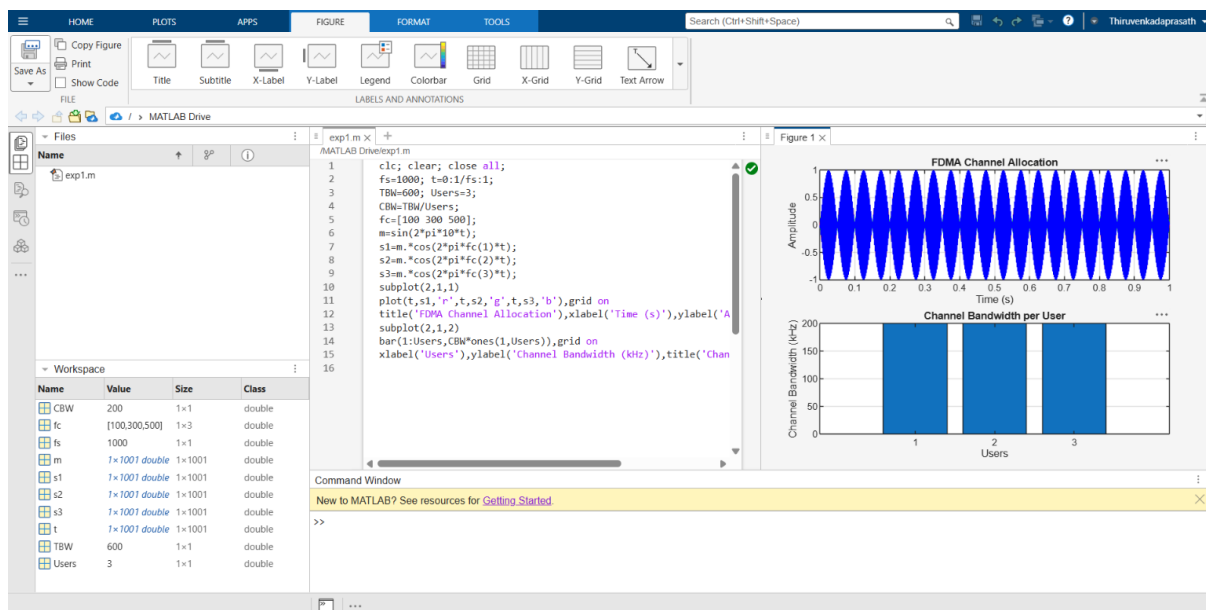
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')

subplot(2,1,2)

bar(1:Users,CBW*ones(1,Users)),grid on

xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

output



Objective 2

Program:

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
```

```
disp(' User Data Rate (kbps)')
```

```
disp([(1:Users)' Rate'])
```

```
figure
```

```
bar(1:Users,Rate)
```

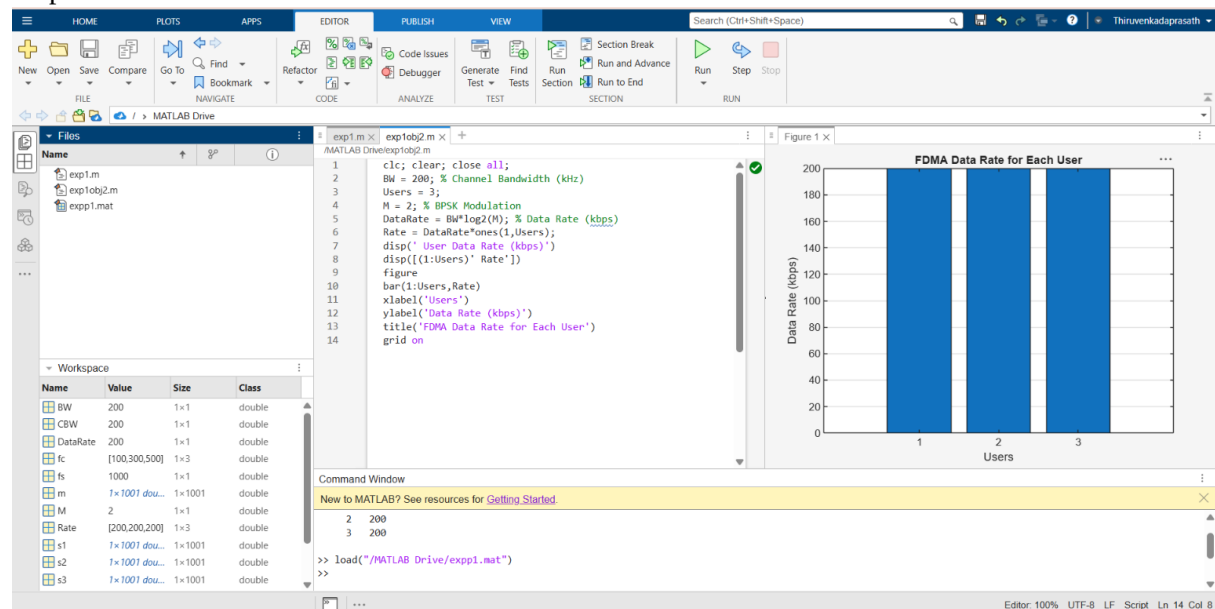
```
xlabel('Users')
```

```
ylabel('Data Rate (kbps)')
```

```
title('FDMA Data Rate for Each User')
```

```
grid on
```

output:



Objective 3:

Program:

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
```

```
disp(' User Data Rate (kbps)')
```

```
disp([ (1:Users)' Rate'])
```

```
figure
```

```
bar(1:Users,Rate)
```

```
xlabel('Users')
```

```
ylabel('Data Rate (kbps)')
```

```
title('FDMA Data Rate for Each User')
```

```
grid on
```

